

Yash Kamble

Game Developer — Gameplay & Systems

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Professional Summary

Solo game developer with 3+ years building complete projects from concept to final build in Unreal Engine and Unity — spanning psychological horror, C++ replay systems, hardware-integrated input, and LLM-driven NPC architecture. Seeking internship or junior gameplay development roles.

Technologies

Engine: Unreal Engine 5 (Blueprints, C++), Unity 2D (C#)

Languages: C++, C#, Python, Arduino (ESP8266)

Tools: Git, Visual Studio, Serial COM Plugin, Audacity, GIMP, Arduino IDE

Platforms: PC, Web

Projects

Time Echo (Unreal Engine · C++)

Portfolio ↑

- TERERecordingComponentBuilt a custom frame-data replay system in C++ —
- captures FFrameData structs (position, rotation, sprint, jump, interact flags) at fixed intervals independent of GPU frame rate, then passes a TArray payload to ATECloneActor for deterministic replay.
- Designed a delegate-driven laser hazard system where clone proximity blocks beams via 3-point height sampling — used as the core puzzle mechanic across 20 handcrafted levels.
- Shipped a polished prototype with modular C++ architecture (recording component, clone actor, interactable base, checkpoint manager) designed for expansion.

Ultra Drive (Unreal Engine · Hardware)

Portfolio ↑

- Built a real-time serial COM pipeline: ESP8266 reads dual ultrasonic sensors + PIR, outputs keyword strings (LEFT / RIGHTDRIFT / RESET), Unreal Engine filters and maps to Blueprint vehicle events at 50ms intervals.
- Implemented distance-based steering depth (5–20 cm → steer, 2–5 cm → drift+brake), giving low-cost digital sensors analog-like expressiveness.

Hello Owner (Unreal Engine · Horror)

Portfolio ↑

- Engineered a rule-based lighting system where player movement during darkness (3s windows) triggers ghost capture and checkpoint respawn — primary fear mechanic, no combat required.
- Built a dynamic rack reconfiguration system using movable actors that shift during lights-off phases to block or reveal maze routes semi-procedurally.
- Designed 3 distinct narrative endings driven by player exploration and decision-making; implemented AI-generated voice via ElevenLabs for solo production.

Dynamic NPC AI System (Unreal Engine · LLM)

Portfolio ↑

- Architected a shared LLM framework where 4 NPCs produce distinct conversational behavior via per-NPC attribute configuration — no scripted dialogue trees, no per-NPC code duplication.
- Implemented free-roaming NPC world behavior with proximity-triggered dialogue interface and stateless text-to-text interaction loop.

Forest of Undead (Unity 2D · Survival Shooter)

itch.io ↑

- Developed an endless survival loop with stat-driven enemy scaling across 5 role-differentiated enemy types and a randomised 3-of-3 upgrade system generating distinct build archetypes each run.

Loop.exe (Unreal Engine · GMTK Jam 2025)

itch.io ↑

- Designed and shipped a memory-execution puzzle with permanent thread loss as the only difficulty axis — built solo in 4 days, submitted 30 hours before deadline.

Education

Terna Engineering College, Navi Mumbai,

Pursuing

Bachelor of Engineering in Information Technology - CGPA: 8.3 / 10